

Mark Scheme — Practice Paper 9 (P1)

M = method mark, A = accuracy mark, B = independent mark. 教师/批改用。

1 $x^2 - 4x - 5 > 0 \Rightarrow (x - 5)(x + 1) > 0$ (M1 A1). $x < -1$ or $x > 5$ (A1). [3]

2 General term $\binom{6}{k} x^{6-k} \left(\frac{2}{x^2}\right)^k = \binom{6}{k} 2^k x^{6-3k}$ (M1). $6 - 3k = 0 \Rightarrow k = 2$ (A1). $\binom{6}{2} \times 2^2 = 15 \times 4$ (M1) = 60 (A1). [4]

3 $2(1 - \sin^2 \theta) = \sin \theta + 1 \Rightarrow 2 \sin^2 \theta + \sin \theta - 1 = 0$ (M1 A1). $(2 \sin \theta - 1)(\sin \theta + 1) = 0$ (M1). $\sin \theta = \frac{1}{2}$: $30^\circ, 150^\circ$; $\sin \theta = -1$: 270° (A1, all three). [4]

4(a) $8 + 8r + 8r^2 = 14 \Rightarrow 8r^2 + 8r - 6 = 0 \Rightarrow 4r^2 + 4r - 3 = 0$ (M1 A1 AG). $(2r - 1)(2r + 3) = 0 \Rightarrow r = \frac{1}{2}$ or $r = -\frac{3}{2}$ (A1). [3]

4(b) Only $r = \frac{1}{2}$ gives $|r| < 1$ (B1). $S_\infty = \frac{8}{1 - \frac{1}{2}} = 16$ (B1). [2]

5(a) Translation by $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (B1); stretch parallel to the y -axis, factor 2 (B1); translation by $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$ (B1).

The stretch must come before the vertical translation. [3]

5(b) $f(x) \geq -5$ (B1). [1]

5(c) $(x - 1)^2 = \frac{y + 5}{2} \Rightarrow x = 1 + \sqrt{\frac{y + 5}{2}}$ (M1, positive root since $x \geq 1$). $f^{-1}(x) = 1 + \sqrt{\frac{x + 5}{2}}$ (A1). [2]

6(a) $(1, 4)$ (B1). [1]

6(b) Gradient = $\frac{6-2}{5+3} = \frac{1}{2}$ (B1). $y - 2 = \frac{1}{2}(x + 3) \Rightarrow x - 2y + 7 = 0$ (B1). [2]

6(c) Gradient $AC \times$ gradient $BC = -1$: $\frac{2}{-3-k} \times \frac{6}{5-k} = -1$ (M1) $\Rightarrow 12 = (k + 3)(5 - k) \Rightarrow k^2 - 2k - 3 = 0$ (A1) $\Rightarrow k = 3$ or $k = -1$ (A1). [Alternative: C lies on the circle with diameter AB : $(k - 1)^2 + 16 = 20$.] [3]

7(a) Minor segment = $\frac{1}{2}(6)^2(2.2 - \sin 2.2) = 18(2.2 - 0.8085) = 25.05$ (M1 A1). Circle area = $36\pi = 113.10$ (B1). Major segment = $113.10 - 25.05 = 88.0 \text{ cm}^2$ (A1; 88.05). [Alternative: sector angle $2\pi - 2.2$ plus triangle: $18(4.083) + 18 \sin 2.2 = 73.5 + 14.55 = 88.0$.] [4]

7(b) Arc = $6 \times 2.2 = 13.2$ (B1). Chord = $2 \times 6 \sin 1.1 = 10.69$ (M1). Perimeter = 23.9 cm (A1). [3]

8(a) $x^2 h = 32 \Rightarrow h = \frac{32}{x^2}$ (B1). $S = x^2 + 4xh = x^2 + 4x \cdot \frac{32}{x^2} = x^2 + \frac{128}{x}$ (B1 AG). [2]

8(b) $\frac{dS}{dx} = 2x - \frac{128}{x^2}$ (B1). $= 0 \Rightarrow x^3 = 64 \Rightarrow x = 4$ (M1 A1). $\frac{d^2S}{dx^2} = 2 + \frac{256}{x^3} = 6 > 0$, so minimum (A1). [4]

8(c) $S = 16 + 32 = 48 \text{ cm}^2$ (B1). [1]

9(a) Roots $x = 0, x = 2$ (B1). $\int_0^2 (3x^2 - 6x) dx = [x^3 - 3x^2]_0^2 = 8 - 12 = -4$ (M1). Area = 4 (A1; must be positive). [3]

$$9(b) \quad V = \pi \int_1^2 \frac{4}{x^2} dx \text{ (M1)} = \pi \left[-\frac{4}{x} \right]_1^2 \text{ (A1)} = \pi(-2 + 4) = 2\pi \text{ (A1)}. [3]$$

$$9(c) \quad \frac{(4x+1)^{5/2}}{\frac{5}{2} \times 4} + c \text{ (M1)} = \frac{1}{10}(4x+1)^{5/2} + c \text{ (A1)}. [2]$$

$$10(a) \quad \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta} = \frac{\sin \theta(1 + \sin \theta) + \cos^2 \theta}{\cos \theta(1 + \sin \theta)} \text{ (M1)} = \frac{\sin \theta + \sin^2 \theta + \cos^2 \theta}{\cos \theta(1 + \sin \theta)} = \frac{1 + \sin \theta}{\cos \theta(1 + \sin \theta)} \text{ (M1)} = \frac{1}{\cos \theta} \text{ (A1 AG)}. [3]$$

$$10(b) \quad \frac{1}{\cos \theta} = \frac{3 \sin \theta}{\cos \theta} \Rightarrow \sin \theta = \frac{1}{3} \text{ (M1 A1)}. \theta = 19.5^\circ, 160.5^\circ \text{ (A1)}. [3]$$

$$10(c) \quad \frac{x}{2} \in [0, \pi]; \tan \frac{x}{2} = -1 \Rightarrow \frac{x}{2} = \frac{3\pi}{4} \text{ (M1)} \Rightarrow x = \frac{3\pi}{2} \text{ (A1; only one solution)}. [2]$$

$$11(a)(i) \quad 20 + 11 \times 3 \text{ (M1)} = 53 \text{ (A1)}. [2]$$

$$11(a)(ii) \quad \frac{n}{2}(40 + 3(n-1)) > 1000 \Rightarrow 3n^2 + 37n - 2000 > 0 \text{ (M1 A1)}. \text{Critical value } n = \frac{-37 + \sqrt{25369}}{6} = 20.4, \text{ so 21 weeks (A1). Check: } S_{20} = 970, S_{21} = 1050. [3]$$

$$11(b) \quad S_n = \frac{3(2^n - 1)}{2 - 1} = 3(2^n - 1) \text{ (M1 A1)}. 3(2^n - 1) > 3000 \Rightarrow 2^n > 1001 \text{ (M1)}. 2^9 = 512, 2^{10} = 1024, \text{ so } n = 10 \text{ (A1)}. [4]$$

$$12(a) \quad OP \perp OQ, \text{ so } PQ \text{ is a diameter (angle in a semicircle) (M1). Centre } (3, 4), \text{ radius} = \frac{1}{2}\sqrt{36 + 64} = 5 \text{ (A1)}. (x-3)^2 + (y-4)^2 = 25 \text{ (A1)}. [\text{Alternative: substitute the three points into } x^2 + y^2 + 2gx + 2fy + c = 0.] [3]$$

$$12(b) \quad \text{Radius } OC \text{ has gradient } \frac{4}{3}, \text{ so tangent gradient } -\frac{3}{4} \text{ (M1)}. y = -\frac{3}{4}x, \text{ i.e. } 3x + 4y = 0 \text{ (A1)}. [2]$$

$$12(c) \quad x = 6: 9 + (y-4)^2 = 25 \Rightarrow (y-4)^2 = 16 \text{ (M1)} \Rightarrow y = 0 \text{ or } 8 \text{ (A1)}. \text{Chord length} = 8 \text{ (A1)}. [3]$$

Total: 75